

# Sunaina Pati

**Website:** sunainapati.github.io

**Current Research Interests:** Complexity Theory, Psuedorandomness, Algebra and Computation and Coding theory.

## EDUCATION

### Chennai Mathematical Institute

India

*BSc (honors) in Computer Science and Mathematics*

*August 2023 – Present (May 2026 expected)*

- Relevant Coursework: Linear Algebra, Group theory, Ring and Field theory, Discrete Mathematics, Probability theory, Design and Analysis of Algorithms, Theory of Computation, Algebra and Computation, Complexity-2, Topics in Number theory, Cryptography, Complexity-1, Algebraic Complexity theory, Algebraic Methods in Theoretical Computer Science, Parallel and Small Space Algorithms, Matroid Theory, Permutation Group Algorithm and Graph Isomorphism, Logic

## RESEARCH EXPERIENCE

### Coding theory and Linear Algebraic Psuedorandomness

August 2026 – present

*under professor Partha Mukhopadhyay and Amit Kumar Sinhababu*

*Chennai Mathematical Institute*

- Reading on Subspace design for my bachelor thesis

### Algebraic Complexity theory

May 2026 – July 2026

*under professor Sébastien Tavenas*

*LAMA Université Savoie Mont Blanc*

- Research funded by RELAX internship program
- Worked Cyclotomic Identity Testing

### Algebraic Complexity theory

May 2025 – July 2025

*under professor Mrinal Kumar*

*Tata Institute of Fundamental Research (TIFR)*

- Reading and researching on the real  $\tau$  conjecture and its application to Algebraic Complexity theory.

## TEACHING ASSISTANT EXPERIENCE AT CHENNAI MATHEMATICAL INSTITUTE

### Graduate Mathematical Toolkit

August 2026 – November 2026

*Under professor Amit Kumar Sinhababu*

### Discrete Mathematics

January 2026 – April 2026

*Under professor V.Arvind*

### Algebra-3(Rings and Field theory)

August 2025 – Present

*Under professor Clare-D-Cruz*

### Algebra-2(Group theory and Advanced Linear algebra)

January 2025 – April 2025

*Under professor Clare-D-Cruz*

### Discrete Mathematics

January 2025 – April 2025

*Under professor V.Arvind and Amit Kumar Sinhababu*

### Algebra-1(Linear algebra)

August 2024 – November 2024

*Under professor Aditya Karnataki*

## HONORS AND AWARDS

### WACT 2026

Copenhagen, 2026

- Attended Workshop in Algebraic Complexity 2026 with travel and stay funded.

### JEST TCS AIR 3

2026

- Recieved AIR 3 in the JEST Theoretical CS examination by IMSC.

### ReLaX internship

2025

- Awarded the ReLaX internship. My research internship with professor Tavenas is funded by ReLaX.

### Preliminary Arizona Winter School: Mathematical cryptography

2025

- Accepted, invited and attending the 9-week school on mathematical cryptography.

### Vigyan Vidushi Math, TIFR

2025

- Accepted, invited and attended to 2-week summer program conducted at Math TIFR, Tata Institute of Fundamental Research.

### Vigyan Vidushi STCS, TIFR

2025

- Accepted, invited and attended to 2-week summer program conducted at STCS TIFR, Tata Institute of Fundamental Research

### SRIRAM Scholarship, CMI

2023 – Present

- Tuition fee waiver for CMI and a monthly stipend for undergraduate studies.

### European Girls' Mathematical Olympiad, Slovenia

2023

- Awarded silver medal at EGMO 2023 held in Slovenia representing India. I scored 29/42 marks.

**Mehta Fellowship, Program in Mathematics for Young Scientists (PROMYS) India, IISC** 2023

- One of the 24 Indian Students accepted ( and attended) the rigorous 6-week program and was awarded Mehta-fellowship.

**Spirit of Ramanujan fellowship** 2022, 2023

- Was awarded Spirit of Ramanujan fellowship ( 6000 dollars each time) in 2023 and 2022 to fund my studies.

**International Mathematical Olympiad Training Camp (IMOTC), HBCSE** 2021, 2022, 2023

- was among the top 30 students in India and was invited for IMO Team selection tests in 2021,2022 and 2023

**European Girls' Mathematical Olympiad Training Camp (EGMOTC), CMI** 2021,2022

- was among the top performing female students in India and was invited for EGMO Team selection tests in 2021 and 2022

## TALKS AND PRESENTATIONS

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**Combinatorial nullstellensatz** 2026

*CMI STEMS Camp* *Chennai Mathematical Institute (CMI)*

- Talk was based on Alon's seminal Combinatorial Nullstellensatz paper. Recording here.

**Cyclic Codes for Error Detection** 2026

*CMI Student Seminar* *Chennai Mathematical Institute (CMI)*

- Talk was based on Peterson and Brown's seminal paper "Cyclic Codes for Error Detection." I also talked about BCH codes and the error correction algorithm. Slides here.

**Primality Testing** 2025

*CMI Student Seminar* *Chennai Mathematical Institute (CMI)*

- Talk was based on Primality testing and talked about the Godel Prize winning paper Primes in P. Write up here.

**On Explicit, almost optimal, epsilon-balanced codes** 2025

*Complexity 2 coursework seminar* *Chennai Mathematical Institute (CMI)*

- Talk was based on Tashma's STOC 2017 Explicit, almost optimal, epsilon-balanced codes for my Complexity-2 coursework presentation.

**On A lower bound for homogeneous algebraic branching programs** 2025

*Algebraic Complexity coursework seminar* *Chennai Mathematical Institute (CMI)*

- Talk was based on Mrinal's CCC A lower bound for homogeneous algebraic branching programs for my Algebraic Complexity coursework presentation. The write up is available here.

## OUTREACH ACTIVITIES

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**Resource Person**

*EGMOTC, INMOTC, IMOTC, PDC* *Chennai Mathematical Institute (CMI)*

- Taught multiple classes on Olympiad geometry and graded the team selection tests at European Girls' Mathematical Olympiad Training Camp (2023,2024), Indian National Mathematics Olympiad training camp of Assam and Tamil Nadu (2025) and International Mathematics Olympiad Training Camp of India (2025).

**Teaching assistant** July 2025

*Awesome Math Summer Program* *Online*

- Graded multiple problems sets daily and conducted problem-solving session every weekday for 3 weeks. Recieved honourium of 2000 dollars.

**Counselor (Teaching Assistant)** May 2024 – June 2024

*PROMYS India* *Indian Institute of Science (IISc)*

- Graded multiple Problem sets on Number theory and mentored students daily. Recieved honourium of 80,000 rs.

**Math Olympiad blog** 2021 – Present

*Sunaina thinks Absurd*

- Math Olympiad blog with over 150k+ views
- Have discussed 500+ problems from various topics

**CMI Student seminar Coordinator** 2024 – 2026

*CMI Student seminar* *Chennai Mathematical Institute (CMI)*

- organising weekly seminars presented by students on various topics